

20260413EB_MN (NCT, ATP)

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2026年4月13日 星期一

split iPSCs into a full 6-well plate on the Friday before differentiation

differentiate 5 of 6-wells with 10mL T2 in cell-repellent plates

keep some iPSCs

2026年4月15日 星期三

T2 -> T3

2026年4月17日 星期五

T3 -> T4 (~14 mL for over-weekend)

2026年4月20日 星期一

T4 -> T5

2026年4月22日 星期三

T5 -> T6

2026年4月24日 星期五

T6 -> T6 (~14mL for over-weekend)

2026年4月27日 星期一

T6 -> T7

Dissociation

1. Collect EBs into 50mL tubes (each plate). Wash the plate with 5mL PBS and added them into the tubes. Let EBs settle with gravity.
2. Remove sup. and add 3mL 0.25% trypsin for each (prepare 16mL 0.25% trypsin with 440U DNaseI)
3. Pipette ~8-10 with P1000 and then incubate the tubes in 37C.
4. pipette*10 at 5min, 10min, and 15min. (for iPSC1000, 1066, and 1108, I stop reaction after I pipette all of them at around 17min. I might overdigest them. I should process it right after or just separately)
5. add 6mL FBS into 50mL tube -> pipette with 5mL pipet ~ 10 times
6. add 6mL CTWM -> pipette ~ 10 times
7. filter cells through 40um cell strainer into new tubes.
8. centrifuge at 200xg, 4min

cell info				^
	A	B	C	
1	cell	passage	well#	
2	WT11	P12	5	

Table1			^
	A	B	
1		volume (mL)	
2	trypsin	4	
3	HI-FBS/CTWM	8	

Table3									/
	A	B	C	D	E	F	G	H	
1	conc. (M/mL)	survival	total volume	tota cell count	plate#	well plate	cell count/well (K)	number of wells	
2	3.35	99	5	16.75	A, B (column1)	24	270	28	
3					B (column2-6)	24	280	20	

Well1							^
	1	2	3	4	5	6	
A	110, 30uL in total						
B							
C	111, 45uL in total						
D							

Well2							^
	1	2	3	4	5	6	
A	vLH185, 2uL/well (8uL in total)	vLH92 (20uL in total) + vLH105 (10uL in total)					
B							
C							
D							

2026年4月28日 星期二

Change T7 -> T7 if added virus

2026年4月30日 星期四

T7 -> BP T8

2026年5月1日 星期五

1/2 BP T8 -> 1/2 BP T8

2026年5月3日 星期日

1/2 BP T8 -> 1/2 BP T8

2026年5月5日 星期二

1/2 BP T8 -> 1/2 BP T8

2026年5月6日 星期三

Imaging: NCT & ATP/ADP with MMS

PlateA							^
	1	2	3	4	5	6	
A	A	B	C	1	2	3	
B	D	E	F	4	5	6	
C	A	B	C	1	2	3	
D	D	E	F	4	5	6	

PlateB ^

	1	2	3	4	5	6
A		A	B	C		
B		D	E	F		
C		2	3	4	7	8
D		5	6	1	9	10

condition ^

	A	B	C	D	E	F	G
1	condition#	gluc conc (mM)	media		condition#	MMS	media
2	1	2.5	no pyruvate or lactate		A	MMS 0mM	with 1mM pyruvate
3	2	0.5	no pyruvate or lactate		B	MMS 1mM	with 1mM pyruvate
4	3	0.15	no pyruvate or lactate		C	MMS 1.5mM	with 1mM pyruvate
5	4	0.05	no pyruvate or lactate		D	MMS 2mM	with 1mM pyruvate
6	5	0.015	no pyruvate or lactate		E	MMS 2.5mM	with 1mM pyruvate
7	6	0	no pyruvate or lactate		F	MMS 3mM	with 1mM pyruvate
8	7	2.5 (1h)	1mM pyruvate				
9	8	0.15 (1h)	1mM pyruvate				
10	9	2.5 (24h)	1mM pyruvate				
11	10	0.15 (24h)	1mM pyruvate				
12							
13							
14							
15							

drug calculation													
	A	B	C	D	E	F	G	H	I	J	K	L	M
1			stock (mM)	1st dilution factor	total volume for 1st dil.	stock volume	media volume for 1st dil.	conc. after 1st dilution	working (mM)	dilution factor from 1st dil.	total volume	volume of 1st dil.	volume of media
2	MM S	A							0		1000	0	1000
3		B	11800	100	100	1	99	118	1	118	1000	8.5	991.5
4		C	11800	100	100	1	99	118	1.5	78.6666666667	1000	12.7	987.3
5		D	11800	100	100	1	99	118	2	59	1000	16.9	983.1
6		E	11800	100	100	1	99	118	2.5	47.2	1000	21.2	978.8
7		F	11800	100	100	1	99	118	3	39.3333333333	1000	25.4	974.6

prepare media with different conc. of glucose with pyruvate							
	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	7	1100	2.5	440	8500	19.3	8480.7
3	8	2.5	0.15	16.6666666667	2000	120.0	1880.0
4				total media volume	10.5	mL	

time table for NCT treatment_MMS



	A	B	C	D	E	F	G
1	NCT						
2	condition#		condition	media	time	imaging	note
3	on 5/6/2026		110				
4	A	A1	0mM MMS	2.5mM gluc	11:03	12:05	
5	B	A2	1mM MMS	2.5mM gluc	11:30	12:40	
6	C	A3	1.5mM MMS	2.5mM gluc	12:03	13:12	
7	D	B1	2mM MMS	2.5mM gluc	12:35	13:45	
8	E	B2	2.5mM MMS	2.5mM gluc	13:10	14:15	XY02 is out of focus for 3rd and 4th frame (if combined)
9	F	B3	3mM MMS	2.5mM gluc	13:42	14:44	
10			111				
11	A	C1	0mM MMS	2.5mM gluc	15:30	16:30	
12	B	C2	1mM MMS	2.5mM gluc	16:03	17:05	
13	C	C3	1.5mM MMS	2.5mM gluc	16:27	17:32	
14	D	D1	2mM MMS	2.5mM gluc	17:03	18:06	XY04 is out of focus at the end
15	E	D2	2.5mM MMS	2.5mM gluc	17:35	18:37	XY01 is out of focus all the time (deleted). XY02 is out of focus at frame 17 of 0009)
16	F	D3	3mM MMS	2.5mM gluc	18:05	19:13	

time table for ATP treatment																								
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V		
1	Perceval sensor			with 1mM pyruvate									with 1mM pyruvate						w/o pyruvate or lactate					
2	#		condition	media	time	imaging on 5/6	neurite		#		condition	treat_time	time	imaging on 5/7	neurite		#		condition	media	time	imaging on 5/8		
3	A	A2	0mM MMS	2.5mM gluc	19:30	20:32	20:47		7	A5	2.5mM gluc w/ 1mM pyruvate	24h	11:00 on 5/6	10:59	11:04		1	C2	2.5	no pyruvate or lactate	18:30	19:30		
4	B	A3	1mM MMS	2.5mM gluc	19:45	20:49	20:53		8	A6	0.15mM gluc w/ 1mM pyruvate	24h	11:00 on 5/6	11:15	11:20		2	C3	0.5	no pyruvate or lactate	18:45	19:45		
5	C	A4	1.5mM MMS	2.5mM gluc	20:03	21:08	21:12		9	B5	2.5mM gluc w/ 1mM pyruvate	1h	10:30	11:34	11:34		3	C4	0.15	no pyruvate or lactate	19:00	20:00		
6	D	B2	2mM MMS	2.5mM gluc	20:20	21:21	21:25		10	B6	0.15mM gluc w/ 1mM pyruvate	1h	10:45	11:45 (This one might have incubator light on) Retook at 11:55			4	D2	0.05	no pyruvate or lactate	19:15	20:15		
7	E	B3	2.5mM MMS	2.5mM gluc	20:40	21:41	21:39										5	D3	0.015	no pyruvate or lactate	19:30	20:30		
8	F	B4	3mM MMS	2.5mM gluc	20:55												6	D4	0	no pyruvate or lactate	19:45	20:45		

O.C. , 60X, zoom1.5, 600x900, 16-bit, Ti2 (new scope)				
	A	B	C	D
1	channel	bit	power (%)	exposure time
2	DIC			
3	488	16	20	200
4	Dapi-ex-Green-em	16	20	300
5	560	16	30	400
6	Dapi-ex-Red-em	16	20	400

NCT_O.C. , 60X, 600x900, 16-bit					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	A1 use 15%
4	488	16	50	1s	

Note: I forgot to add SirDNA for A1 and A2, so I add 0.2uL directly to the well at 11:40.

fileID_treatment_NCT					
	A	B	C	D	E
1	image	reporter	treatment	time	pyruvate
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	0mM MMS	1h	yes
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	0mM MMS	1h	yes
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	1mM MMS	1h	yes
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	1mM MMS	1h	yes
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	1.5mM MMS	1h	yes
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	1.5mM MMS	1h	yes
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	2mM MMS	1h	yes
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	2mM MMS	1h	yes
10	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	2.5mM MMS	1h	yes
11	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	2.5mM MMS	1h	yes
12	com_110_MN_no-glu_50%_1s_30s_60x_0010	110	3mM MMS	1h	yes
13	com_110_MN_no-glu_50%_1s_30s_60x_0011	110	3mM MMS	1h	yes
14	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	0mM MMS	1h	yes
15	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	0mM MMS	1h	yes
16	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	1mM MMS	1h	yes
17	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	1mM MMS	1h	yes
18	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	1.5mM MMS	1h	yes
19	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	1.5mM MMS	1h	yes
20	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	2mM MMS	1h	yes
21	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	2mM MMS	1h	yes
22	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	2.5mM MMS	1h	yes
23	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	2.5mM MMS	1h	yes
24	com_111_MN_no-glu_50%_1s_30s_60x_0010	111	3mM MMS	1h	yes
25	com_111_MN_no-glu_50%_1s_30s_60x_0011	111	3mM MMS	1h	yes

fileid_treat_20260506_MN_92

	A	B	C	D
1	image	treatment	treat_time	pyruvate
2	vLH92_MN_60x-1xzoom_0000	0mM MMS	1h	yes
3	vLH92_MN_60x-1xzoom_0001	1mM MMS	1h	yes
4	vLH92_MN_60x-1xzoom_0002	1.5mM MMS	1h	yes
5	vLH92_MN_60x-1xzoom_0003	2mM MMS	1h	yes
6	vLH92_MN_60x-1xzoom_0004	2.5mM MMS	1h	yes
7	vLH92_MN_60x-1xzoom_0005	3mM MMS	1h	yes
8	neurite_vLH92_MN_60x-1xzoom_000	0mM MMS	1h	yes
9	neurite_vLH92_MN_60x-1xzoom_001	1mM MMS	1h	yes
10	neurite_vLH92_MN_60x-1xzoom_002	1.5mM MMS	1h	yes
11	neurite_vLH92_MN_60x-1xzoom_003	2mM MMS	1h	yes
12	neurite_vLH92_MN_60x-1xzoom_004	2.5mM MMS	1h	yes
13	neurite_vLH92_MN_60x-1xzoom_005	3mM MMS	1h	yes

2026年5月7日 星期四

- Cells with MMS from yesterday died. (A1 also died though it has no MMS. I suspect it was because the SirDNA)

Imaging: ATP/ADP with 1h and 24h glucose deprivation

fileid_treat_20260507_MN_92

	A	B	C	D	E
1	image	treatment	treat_time	pyruvate	
2	vLH92_MN_60x-1xzoom_0000	2.5mM gluc	24h	yes	
3	vLH92_MN_60x-1xzoom_0001	0.15mM gluc	24h	yes	
4	vLH92_MN_60x-1xzoom_0002	2.5mM gluc	1h	yes	
5	vLH92_MN_60x-1xzoom_0003	0.15mM gluc	1h	yes	
6	vLH92_MN_60x-1xzoom_0004	0.15mM gluc	1h	yes	same position as 0003 (retook because the light was on)
7	neurite_vLH92_MN_60x-1xzoom_000	2.5mM gluc	24h	yes	
8	neurite_vLH92_MN_60x-1xzoom_001	0.15mM gluc	24h	yes	
9	neurite_vLH92_MN_60x-1xzoom_002	2.5mM gluc	1h	yes	
10	neurite_vLH92_MN_60x-1xzoom_003	0.15mM gluc	1h	yes	
11	neurite_vLH92_MN_60x-1xzoom_004	0.15mM gluc	1h	yes	same position as 0003 (retook because the light was on)

2026年5月8日 星期五

Imaging: NCT and ATP/ADP with [glucose] titration

Treat SirDNA 1:6000 (1uL in 20uL (20x dilution), 3.3uL/mL)

Note: I found the water bath dried out at the end, but not sure about when. I am sure there was water at the beginning.

prepare media with different conc. of glucose

	A	B	C	D	E	F	G
1		diluted from (mM)	targetconc. (mM)	dilution factor	final volume (uL)	added volume (uL)	media volume
2	1	1100	2.5	440	6000	13.6	5986.4
3	2	2.5	0.5	5	4000	800.0	3200.0
4	3	2.5	0.15	16.6666666667	4000	240.0	3760.0
5	4	2.5	0.05	50	4000	80.0	3920.0
6	5	2.5	0.015	166.6666666667	4000	24.0	3976.0
7	6		0		5000		1144.0
8				total media volume	27	mL	

time table for NCT treatment_glucose 5/8



	A	B	C	D	E	F	G
1	NCT						
2	condition#		condition	media	time	imaging	note
3	on 5/6/2026		110				
4	1	A4	2.5	no pyruvate or lactate	11:30	12:34	
5	2	A5	0.5	no pyruvate or lactate	12:02	13:06	
6	3	A6	0.15	no pyruvate or lactate	12:30	13:45	
7	4	B4	0.05	no pyruvate or lactate	13:03	14:15	
8	5	B5	0.015	no pyruvate or lactate	14:33	15:31	
9	6	B6	0	no pyruvate or lactate	15:05	16:01	
10			111				
11	1	C4	2.5	no pyruvate or lactate	15:30	16:36	
12	2	C5	0.5	no pyruvate or lactate	16:00	17:05	
13	3	C6	0.15	no pyruvate or lactate	16:32	17:33	
14	4	D4	0.05	no pyruvate or lactate	17:00	18:02	XY01 is not focused 2nd and 3rd frame of 0006 and the first frame of 0007
15	5	D5	0.015	no pyruvate or lactate	17:30	18:39	bubble delay
16	6	D6	0	no pyruvate or lactate	18:00	19:05	

time table for ATP treatment_5/8									
	A	B	C	D	E	F	G	H	I
1	w/o pyruvate or lactate								
2	#		conditio n	media	time	imagin g on 5/8			
3	2	C2	0.5	no pyruvate or lactate	18:50	19:45			
4	3	C3	0.15	no pyruvate or lactate	19:05	20:13			
5	4	C4	0.05	no pyruvate or lactate	19:20	20:26			
6	5	D2	0.015	no pyruvate or lactate	19:35	20:35			
7	6	D3	0	no pyruvate or lactate	20:00	21:01			
8	1	D4	2.5	no pyruvate or lactate	20:16	21:17			
9		D4	2.5	calibration		21:29- 21:50	use 10 points from above		

NCT_O.C. , 60X, 600x900, 16-bit2					
	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	30	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	

pHred reference: <https://pubs.acs.org/doi/10.1021/ja202902d>

pH correction from the paper				^
	A	B	C	
1	Time	Condition	Purpose	
2	0–5 min	Baseline, 2.5 mM glucose	Establish stable ATP/pH baseline	
3	5–8 min	5–10 mM NH ₄ Cl + 2.5 mM glucose	pH-only calibration pulse	
4	8–15 min	Washout, 2.5 mM glucose	Recovery from acid load	
5	15–45 min	0 mM glucose	Measure ATP:ADP drop and pH change	
6	45–48 min	Optional NH ₄ Cl + 0 mM glucose	Check low-ATP pH contour	
7	48–60 min	Washout, 0 mM glucose	Recovery behavior under low ATP	
8	60–80 min	2.5 mM glucose refeeding	ATP recovery	

stock: 1.43M NH4Cl
working: 10mM NH4Cl
--> prepare 10mM directly to media (142x), 1mL: 993uL media + 7uL 1.43M NH4Cl

First trial of pH correction. Image right after finished D4 imaging.
NH4Cl short pulse after imaging: use D4 (with 2.5mM glucose)
baseline: for 5min, image every 20 s.
10mM NH4Cl pulse: 3 min, image every 20 s.
2.5mM gluc recovery: 7min, image every 20s

fileID_treatment_20260508_MN_110,111

	A	B	C	D	E	F
1	image	reporter	treatment	time	pyruvate	
2	com_110_MN_no-glu_50%_1s_30s_60x_0000	110	2.5mM gluc	1h	no	
3	com_110_MN_no-glu_50%_1s_30s_60x_0001	110	2.5mM gluc	1h	no	
4	com_110_MN_no-glu_50%_1s_30s_60x_0002	110	0.5mM gluc	1h	no	
5	com_110_MN_no-glu_50%_1s_30s_60x_0003	110	0.5mM gluc	1h	no	
6	com_110_MN_no-glu_50%_1s_30s_60x_0004	110	0.15mM gluc	1h	no	
7	com_110_MN_no-glu_50%_1s_30s_60x_0005	110	0.15mM gluc	1h	no	
8	com_110_MN_no-glu_50%_1s_30s_60x_0006	110	0.05mM gluc	1h	no	
9	com_110_MN_no-glu_50%_1s_30s_60x_0007	110	0.05mM gluc	1h	no	
10	com_110_MN_no-glu_50%_1s_30s_60x_0008	110	0.015mM gluc	1h	no	
11	com_110_MN_no-glu_50%_1s_30s_60x_0009	110	0.015mM gluc	1h	no	
12	com_110_MN_no-glu_50%_1s_30s_60x_0010	110	0mM gluc	1h	no	
13	com_110_MN_no-glu_50%_1s_30s_60x_0011	110	0mM gluc	1h	no	
14	com_111_MN_no-glu_50%_1s_30s_60x_0000	111	2.5mM gluc	1h	no	
15	com_111_MN_no-glu_50%_1s_30s_60x_0001	111	2.5mM gluc	1h	no	
16	com_111_MN_no-glu_50%_1s_30s_60x_0002	111	0.5mM gluc	1h	no	
17	com_111_MN_no-glu_50%_1s_30s_60x_0003	111	0.5mM gluc	1h	no	
18	com_111_MN_no-glu_50%_1s_30s_60x_0004	111	0.15mM gluc	1h	no	
19	com_111_MN_no-glu_50%_1s_30s_60x_0005	111	0.15mM gluc	1h	no	
20	com_111_MN_no-glu_50%_1s_30s_60x_0006	111	0.05mM gluc	1h	no	XY01 is not focused 2nd and 3rd frame of 0006 and the first frame of 0007
21	com_111_MN_no-glu_50%_1s_30s_60x_0007	111	0.05mM gluc	1h	no	
22	com_111_MN_no-glu_50%_1s_30s_60x_0008	111	0.015mM gluc	1h	no	
23	com_111_MN_no-glu_50%_1s_30s_60x_0009	111	0.015mM gluc	1h	no	
24	com_111_MN_no-glu_50%_1s_30s_60x_0010	111	0mM gluc	1h	no	
25	com_111_MN_no-glu_50%_1s_30s_60x_0011	111	0mM gluc	1h	no	

fileid_treat_20260508_MN_92

	A	B	C	D
1	image	treatment	treat_time	pyruvate
2	vLH92_MN_60x-1xzoom_0000	0.5mM gluc	1h	no
3	vLH92_MN_60x-1xzoom_0001	0.15mM gluc	1h	no
4	vLH92_MN_60x-1xzoom_0002	0.05mM gluc	1h	no
5	vLH92_MN_60x-1xzoom_0003	0.015mM gluc	1h	no
6	vLH92_MN_60x-1xzoom_0004	0mM gluc	1h	no
7	vLH92_MN_60x-1xzoom_0005	2.5mM gluc	1h	no
8	vLH92_MN_60x-1xzoom_0006	2.5mM gluc	5min	no
9	vLH92_MN_60x-1xzoom_0007	NH4Cl	3min	no
10	vLH92_MN_60x-1xzoom_0008	2.5mM gluc recover	7min	no
11	neurite_vLH92_MN_60x-1xzoom_000	0.5mM gluc	1h	no
12	neurite_vLH92_MN_60x-1xzoom_001	0.15mM gluc	1h	no
13	neurite_vLH92_MN_60x-1xzoom_002	0.05mM gluc	1h	no
14	neurite_vLH92_MN_60x-1xzoom_003	0.015mM gluc	1h	no
15	neurite_vLH92_MN_60x-1xzoom_004	0mM gluc	1h	no
16	neurite_vLH92_MN_60x-1xzoom_005	2.5mM gluc	1h	no

2026年5月9日 星期六

Imaging: non-classical import rate

time table for NCT treatment_20260509_MN_185

	A	B	C	D	E	F	G
1	NCT						
2	condition#		condition	media	time	imaging	note
3	on 5/9/2026		185				
4	1	A1	2.5	no pyruvate or lactate	15:45	16:44	
5	3	B1	0.15	no pyruvate or lactate	16:15	17:20	
6	5	C1	0.015	no pyruvate or lactate	16:43	17:49	
7	6	D1	0	no pyruvate or lactate	17:15	18:24	

NCT_O.C. , 60X, 600x900, 16-bit1

	A	B	C	D	E
1	channel	bit	power (%)	exposure time	note
2	561	16	35	300 ms	
3	640	16	10	100 ms	
4	488	16	50	1s	

fileID_treatment_20260509_MN_185						
	A	B	C	D	E	F
1	image	reporter	treatment	time	pyruvate	
2	com_185_MN_no-glu_50%_1s_30s_60x_0000	185	2.5mM gluc	1h	no	
3	com_185_MN_no-glu_50%_1s_30s_60x_0001	185	2.5mM gluc	1h	no	
4	com_185_MN_no-glu_50%_1s_30s_60x_0002	185	0.15mM gluc	1h	no	
5	com_185_MN_no-glu_50%_1s_30s_60x_0003	185	0.15mM gluc	1h	no	out of focus (a bit): XY4, frame 19, 28-30 XY5: 28 XY6: 28
6	com_185_MN_no-glu_50%_1s_30s_60x_0004	185	0.015mM gluc	1h	no	
7	com_185_MN_no-glu_50%_1s_30s_60x_0005	185	0.015mM gluc	1h	no	
8	com_185_MN_no-glu_50%_1s_30s_60x_0006	185	0mM gluc	1h	no	
9	com_185_MN_no-glu_50%_1s_30s_60x_0007	185	0mM gluc	1h	no	position5, frame26 out of focus,